



FUNDAMENTAL OF SOFTWARE SECURITY

Alemu Woldie

awcourseapp@gmail.com
Software Engineering
Wolkite, Ethiopia

Chapter Two - Secure System Design

After completing this chapter, You're expected to:

- Know more about Core Security Requirements and their respective definitions
- Assess risk, analyze the risk and take the respective mitigation techniques
- Know about Reduction of Risk Surface, Threat Modeling, Development Methodologies, Architecture and Secure Technologies
- Tenets of Secure Architecture and Design

Model for Network Security

- A message is to be transferred from one party to another across some sort of Internet service.
- The two parties must cooperate for the exchange to take place.
- A logical information channel is established by defining routes & communication protocols by the principals.
- Security aspects come into play when it is necessary to protect the information transmission from an opponent.
- All the techniques for providing security have two components:
 - A security-related transformation on the information to be sent
 - Some secret information shared by the two principals
- A general model for Network Security is presented in figure 1

Model for Network Security

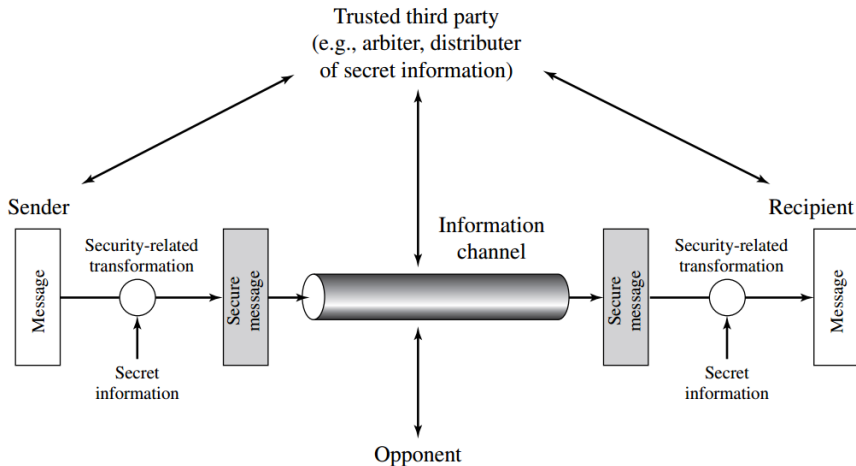


Figure 1: Model for Network Security

Model for Network Security

- A trusted third party may be needed to achieve secure transmission
 - Responsible for distributing the secret information to the two principals while keeping it from any opponent.
 - Arbitrate disputes between the two principals concerning the authenticity of a message transmission.
- The general model shows that there are four basic tasks in designing a particular security service:
 - Design an algorithm for performing the security-related transformation.
 - Generate the secret information to be used with the algorithm.
 - Develop methods for the distribution and sharing of the secret information.
 - Specify protocols to be used by the principals that makes use of the security algorithm & secret information.

Model for Network Security

- There are other security-related situations of interest that do not neatly fit this model
- A general model of these other situations is illustrated in figure 2
- The model reflects a concern for protecting an information system from unwanted access.

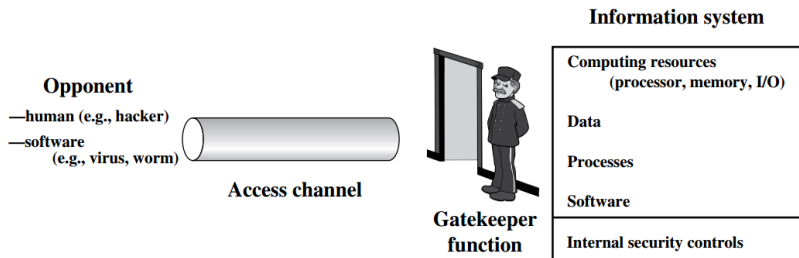


Figure 2: Network Access Security Model

Model for Network Security

- Most users are familiar with the concerns caused by the existence of hackers, who attempt to penetrate systems that can be accessed over a network.
- The hacker can be someone who, with no malign intent, simply gets satisfaction from breaking and entering a computer system.
- The intruder can be a disgruntled employee who wishes to do damage or a criminal who seeks to exploit computer assets for financial gain.
- Another type of unwanted access is the placement in a computer system of logic that exploits vulnerabilities in the system and that can affect application programs as well as utility programs.

Model for Network Security

- The security mechanisms needed to cope with unwanted access fall into two broad categories:
 - The first category might be termed a gatekeeper function. It includes password-based login procedures that are designed to deny access to all but authorized users and screening logic that is designed to detect and reject worms, viruses, and other similar attacks.
 - The second line of defense consists of a variety of internal controls that monitor activity and analyze stored information in an attempt to detect the presence of unwanted intruders.

Network Security Design Best Practices and Principles

- With all the variations of network security design, some important questions surface, including the following:
 - Identifying methods to reasonably secure components
 - Knowing when you've done enough to lock things down
 - Selecting the best design to maximize resilience
- There is no single answer will suffice the above issues, but there are right ways and wrong ways to integrate security into a network.
- Regardless of the size/complexity of your network, three main factors constitute a secure & resilient network:
 - What you have
 - Where it's located
 - How and when it's managed

Network Security Design Best Practices and Principles

- Here are some basic practices and principles for building network security systems:
 - Understanding network threats
 - Doing what is essential but nothing more
 - Do vulnerability assessment
 - Follow layering approach
 - Use well defined trust models
 - Implement service operations
- *Basic references:*
- SearchNetworking. 2022. Network security design best practices and principles: Keep it simple. [online] Available at: <https://www.techtarget.com/searchnetworking/tip/Network-security-design-best-practices-and-principles-Keep-it-simple> [Accessed 17 April 2022].
- www.ncsc.gov.uk. (n.d.). Cyber security design principles. [online] Available at: <https://www.ncsc.gov.uk/collection/cyber-security-design-principles/cyber-security-design-principles>.